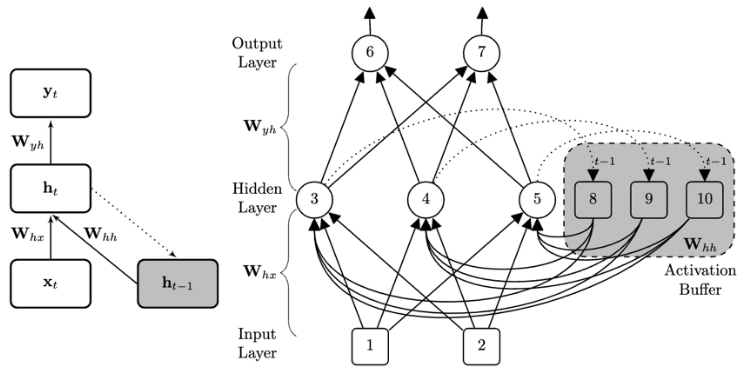
SYSC 5108 Winter 2026 Assignment 4

1. Assume a simple recurrent neural network architecture as shown below.



This network has two input neurons, three ReLUs in the hidden layer, and two ReLUs in the output layer. Also, all the bias terms in the network are equal to 0.1, and the weight matrices of the network (excluding bias terms) are listed below.

$$\underbrace{\begin{bmatrix} -0.07 & 0.05 \\ -0.04 & 0.1 \\ -0.05 & -0.05 \end{bmatrix}}_{\mathbf{W}_{hx}} \underbrace{\begin{bmatrix} -0.22 & -0.1 & 0.05 \\ -0.04 & -0.09 & -0.06 \\ -0.09 & 0 & -0.08 \end{bmatrix}}_{\mathbf{W}_{hh}} \underbrace{\begin{bmatrix} 0.06 & -0.01 & -0.18 \\ -0.06 & 0.13 & 0.14 \end{bmatrix}}_{\mathbf{W}_{yh}}$$

   a. If $\boldsymbol{x}_t = [1, 0.5]$ and $\boldsymbol{h}_{t-1} = [0.05, 0.2, 0.15]$, caculate the value of $\boldsymbol{y}_t$.
   b. Assuming that the target output at time t be [0, 1], calculate the δ value for each neuron in the network.

2. Consider the example shown in Slide 10 of Chapter 6. Let n represent the time starting from 0 when the sender is ready to send a message.
   a. Find $\pi_0(n)$ and $\pi_1(n)$.
   b. Find the probability that the sender will revisit state 0.
   c. Find the time average probabilities of the sender in state 0, $\bar{\pi}_0$ and in state 1, $\bar{\pi}_1$.
   d. Find the probability that the sender is in state 0 before the message is sent successfully.

3. Let the transition probability matrix of a two-state Markov chain be given by

$$P = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$

   a. Suppose at time 0, the distribution of states is $Q_0 = [0.2 \quad 0.8]^{\intercal}$. Find the marginal distribution Q_1 at time 1.
   b. Show that

$$P^{(n)} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2}(2p-1)^n & \frac{1}{2} - \frac{1}{2}(2p-1)^n \\ \frac{1}{2} - \frac{1}{2}(2p-1)^n & \frac{1}{2} + \frac{1}{2}(2p-1)^n \end{bmatrix}$$

c. Find the limiting probabilities.
d. Also find the result for the following operation

$$\lim_{n \to \infty} P^{(n)}$$

4. In the grid world example, rewards are positive for the gold state, negative for traps, zero for others. Are the absolute values of these rewards important, or only the intervals between them? Prove that adding a constant c to all the rewards adds a constant, v_c, to the values of all states, and thus does not affect the relative values of any states under any policy. What is v_c in terms of c and γ?

5. Consider a two-state-two-action reinforcement learning scenario with the following environment.

State	Action	Next State	Probability	Reward
0	0	0	1	-1
0	1	1	1	1
1	0	1	1	1
1	1	0	1	-1

Assume $\gamma = 0.1$ and an ε-greedy policy with $\pi(1|0) = \pi(0|1) = 0.6$.

a. Find the value of ε.
b. Find the state-value function and action-value function using Bellman equation.
c. Find the optimal policy with Bellman equations.
d. With the given ε-greedy policy, starting from state 0 and 1 at time 0 respectively, generate two trajectories using the following random number sequence and terminate at $t = 3$. Assume $\alpha = 0.6$, estimate the state-value function and action-value function with Monte-Carlo method using the generated trajectories.
0.94 0.27 0.30 0.18 0.75 0.35 0.61 0.42 0.48 0.26 0.88 0.31 0.90 0.55 0.95
e. Assume $\alpha = 0.6$, starting with all action values equal to 0, find action values using off-policy TD control using the trajectories generated in (d).
f. Assume $\alpha = 0.6$, starting with all action values equal to 0 and an initial policy $\pi(1|0) = \pi(0|1) = 0.5$, estimate the action-value function using on-policy control with one trajectory starting with state 0 and ending at $t = 5$. Use the same random sequence as in (d) and the same ε-value as in (a).

6. Given the following two-dimensional dataset. Find a) the one-dimensional PCA; b) the encoded points of the one-dimensional PCA; c) the reconstructed points; d) the percentage of variance explained, i.e., the variance ratio of encoded points and original data points.

x_1	2	0	-3	3	-2
x_2	-2	4	-1	-3	2

7. We have the following 5 data points (observed values in 1D): x = [1, 2, 3, 9, 10]. Assume a 2-component Gaussian Mixture Model: $p(x) = \pi_1 \cdot \mathcal{N}(x \mid \mu_1, \sigma_1^2) + \pi_2 \cdot \mathcal{N}(x \mid \mu_2, \sigma_2^2)$, where $\pi_1 + \pi_2 = 1$, and we assume spherical/unit variance for simplicity: $\sigma_1^2 = \sigma_2^2 = 1$ (fixed — we will only estimate π, μ_1, μ_2). Initial parameter values: $\pi_1 = 0.6$, $\pi_2 = 0.4$, $\mu_1 = 2.0$, $\mu_2 = 9.5$. What are the updated parameter values after one EM iteration? (You can stop after one iteration — no need to iterate until convergence.)

8. Consider a two-dimensional RV $X = (X_0, X_1)$ with each X_i being binary, i.e., $X_i \in \{0, 1\}$. The joint probabilities of the two RVs are as follows.

$$P(X_0, X_1) = \begin{pmatrix} \frac{1}{4} & \frac{1}{8} \\ \frac{1}{2} & \frac{1}{8} \end{pmatrix}$$

   a. What will be the distribution of $P(X_0|X_1)$ and $P(X_1|X_0)$
   b. Run Gibbs sampling for 3 iterations starting with uniform distributions for both X_0 and X_1 independently with the following sequence of random numbers.
   0.94 0.27 0.30 0.18 0.75 0.35 0.61 0.42 0.48 0.26 0.88 0.31 0.90 0.55 0.95